

JAVA SDE-2 INTERVIEW PREPARATION SERIES

SYSTEM DESIGN AND BACKEND - BOOK 04 OF 14 - STUDY STEP 19E

Spring Framework for Java Engineers

IoC, Dependency Injection, Beans, AOP, Transactions, and Testing Foundations

BY

Vinay Reddy Kalluri

A fundamentals-first roadmap for the Spring container and the runtime mechanics that Spring Boot builds upon.

SPRING | IOC | DI | BEANS | AOP | TRANSACTIONS

JAVA 21 | INTERVIEW CORE | SDE-2 FOLLOW-UPS | PRINTABLE EDITION

Series edition - Release 2026-07-29

Open educational interview-preparation edition

Start Here - Your Route Through This Volume

60-second entry rule: If two or more recognition signals below expose a gap, begin with Chapter 1. If the prerequisites are weak, step back to SD 03 – Hibernate and JPA. If you can already explain every outcome without notes, jump to Part II and prove it through the practice ladder.

Make core Spring behavior explicit before readers use Spring Boot conventions and automation.

Readiness map

Decision	Evidence
START HERE	Object construction and wiring are container-managed; Proxy boundaries change method behavior; A transaction or lifecycle annotation has runtime consequences.
PREPARE FIRST	Java Engineering foundations; Hibernate and JPA Book 03 recommended.
MOVE ON WHEN	Understand the planned Spring Framework progression; Distinguish container contracts from conveniences; Know the core mechanics expected in SDE-2 interviews.

Choose the route that fits today

- 1 **Learning:** read in order, type the representative Java examples, and complete each dry run.
- 2 **Revision or rehearsal:** start at the first decision map, invariant, or failure mode you cannot explain; if none expose a gap, go directly to Part II and prove readiness without notes.

System Design Segment Position

SD 03 - Hibernate and JPA	YOU ARE HERE SD 04 - Spring Framework	SD 05 - Spring Boot
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Part I - Core Learning**SDE-2 CORE CHAPTER**

Chapter 1 - Spring Framework Foundations - Planned Learning Roadmap

Publication status: roadmap edition. Later revisions will add runnable container examples, proxy traces, test cases, and interview exercises.

Spring Framework is the foundation beneath Spring Boot and the wider Spring ecosystem. This book will teach the container, dependency injection, application context, proxy model, transactions, events, validation, and web foundations before convenience auto-configuration is introduced.

Planned sequence

- 1 Inversion of control, dependency injection, object graphs, and explicit boundaries.
- 2 Bean definitions, component scanning, configuration classes, scopes, and lifecycle.
- 3 Application contexts, environment properties, profiles, resources, and events.
- 4 Aspect-oriented programming, proxies, advice, self-invocation, and visibility limits.
- 5 Declarative transactions, propagation, isolation, rollback rules, and proxy boundaries.
- 6 Validation, conversion, formatting, data binding, and message resolution.
- 7 Spring MVC request flow, controllers, exception handling, and HTTP contracts.
- 8 Testing the container with focused slices and explicit integration boundaries.

Interview focus

Readers will learn to explain what the container creates, which calls pass through a proxy, why a transaction annotation may not take effect, how bean scope changes ownership, and when constructor injection improves correctness. The book will keep interface contracts separate from framework magic.

Completion gate

A reader is ready for Spring Boot when they can construct and test a small Spring application context, reason about bean lifecycle and proxy interception, define transaction boundaries, and diagnose configuration without relying on trial-and-error annotations.

Part II - Practice and Handoff

Practice Ladder and Next Steps

TRY IT | Practice ladder

- Foundation: IoC, beans, and injection
- Interview Core: scopes, proxies, and transactions
- SDE-2 Follow-up: extension points and test design

For every coding problem, state the input contract, select the numeric and collection types deliberately, write the invariant before the loop or recursion, test the smallest and largest valid inputs, and close with time and auxiliary-space complexity.

Navigation

[Previous book: SD 03 - Hibernate and JPA for Java Backend Interviews](#)

[Series index](#)

[Next book: SD 05 - Spring Boot for Java Backend Engineers](#)

TRY IT | Completion check

- ☐ I can recognize the core patterns without being told the data structure or algorithm.
- ☐ I can implement the representative Java templates from memory.
- ☐ I can explain why each implementation is correct.
- ☐ I can test boundaries, invalid inputs, overflow, and adversarial shapes.
- ☐ I can answer at least one SDE-2 follow-up involving trade-offs or production constraints.

Series Roadmap

Choose Java, DSA, or System Design and Backend first, then follow the books inside that segment in order. Stable study-step codes remain on filenames and links; the clearer segment book codes appear on the website and every PDF cover.

SEGMENT	BOOK	FOCUSED LEARNING PATH
Java	JAVA 01	Java Foundations for Problem Solving
Java	JAVA 02	Git and GitHub for Java Engineers
Java	JAVA 03	Maven and Gradle for Java Engineers
Java	JAVA 04	Advanced Java B: Language, OOP, and Modern Java
Java	JAVA 05	Advanced Java C: Collections, Streams, and I/O
Java	JAVA 06	Advanced Java A: JVM and Execution
Java	JAVA 07	Advanced Java D: Concurrency and the Memory Model
Java	JAVA 08	Advanced Java E: Performance, Diagnostics, and GC Incidents
Java	JAVA 09	Advanced Java G: Question Bank, Study Plan, and Reference
DSA	DSA 01	Time and Space Complexity for Java Interviews
DSA	DSA 02	Number Systems and Math Foundations for DSA Interviews
DSA	DSA 03	Number Systems Interview Patterns and Rapid Revision
DSA	DSA 04	Bit Manipulation in Java
DSA	DSA 05	Loop Mastery, Patterns, and Index Calculations
DSA	DSA 06	Arrays and Array Problem-Solving Patterns
DSA	DSA 07	Strings and String Problem-Solving Patterns
DSA	DSA 08	Hashing: Maps, Sets, Frequency, and Prefix State
DSA	DSA 09	Recursion and Backtracking in Java
DSA	DSA 10	Linked Lists: Pointer Reasoning and Mutation
DSA	DSA 11	Stacks, Queues, Deques, and Monotonic Patterns
DSA	DSA 12	Binary Search: Bounds, Answers, and Invariants
DSA	DSA 13	Trees, Binary Search Trees, and Tries
DSA	DSA 14	Heaps, Priority Queues, Selection, and Top-K
DSA	DSA 15	Graphs: Traversal, Ordering, Paths, and Union-Find
DSA	DSA 16	Greedy Algorithms: Recognition, Proofs, and Scheduling
DSA	DSA 17	Dynamic Programming: State, Transitions, and Optimization
System Design	SD 01	Advanced Java F: Design, Backend, Testing, and Security
System Design	SD 02	MySQL for Java Backend Interviews
System Design	SD 03	Hibernate and JPA for Java Backend Interviews
System Design	SD 04	Spring Framework for Java Engineers
System Design	SD 05	Spring Boot for Java Backend Engineers

SEGMENT	BOOK	FOCUSED LEARNING PATH
System Design	SD 06	Spring Data for Java Backend Engineers
System Design	SD 07	MongoDB for Java Backend Interviews
System Design	SD 08	Redis for Java Backend Interviews
System Design	SD 09	Apache Kafka and Spring Kafka
System Design	SD 10	Spring Ecosystem Extensions
System Design	SD 11	Spring AI for Java Engineers
System Design	SD 12	Advanced Java H: Spring Boot and REST APIs
System Design	SD 13	Advanced Java I: Persistence, SQL, JPA, and Caching
System Design	SD 14	Advanced Java J: Distributed Systems and System Design

Roadmap editions identify planned books whose chapter sets will be expanded one at a time. Existing deep-dive books remain available later in each segment, so no published material is displaced.

About the Author

Vinay Reddy Kalluri

Vinay Reddy Kalluri is a senior Java backend engineer, the creator and founding author of the Java SDE-2 Interview Preparation Series, and its Editor-in-Chief and Chief Auditor. His work spans high-throughput microservices, event-driven architecture, resilient APIs, large-scale data processing, cloud delivery, and production performance across healthcare and enterprise systems.

Editorial leadership

As Editor-in-Chief, Vinay owns the learning sequence, scope, voice, and publication decisions. As Chief Auditor, he owns the Java accuracy standard, validation evidence, attribution review, and release-readiness gate. Individual contributors retain visible credit for accepted original work through AUTHORS.md and the public Git history.

What distinguishes his approach is the combination of hands-on system design and operational ownership. He treats timeouts, retries, idempotency, dead-letter recovery, data consistency, SQL behavior, caching, and observability as part of the design rather than afterthoughts. He has also mentored engineers and led modernization, migration, and reliability work across distributed services.

Selected engineering and academic highlights

- More than eight years building Java and Spring Boot services for high-throughput healthcare and enterprise platforms.
- Designed Kafka-based event systems that have processed more than one million events per hour, with explicit idempotency, retry, recovery, and observability controls.
- M.S. in Computer Science from the University of Missouri-Kansas City (3.9/4.0) and M.S. in Software Engineering from VIT University (9.3/10), where he was a Gold Medalist.
- Independent builder of Work Visa Insights, an immigration-data intelligence platform, and Play Together, a published Android application.

Why this series exists

Vinay created this series to turn fragmented interview preparation into a deliberate progression: learn the fundamentals, run the examples, predict behavior, debug failures, explain trade-offs, and only then move into advanced engineering. The books reflect the same principle he applies to production systems: clarity before cleverness, explicit contracts, measurable behavior, and reliable foundations.

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Repository: <https://github.com/vinayreddykalluri/SDE2-Interview-Handbook>. Attribution does not imply endorsement. Java and related marks are owned by their respective holders.

Examples target Java 21 unless a section states otherwise. Validate release-specific APIs, JVM behavior, security, performance, licensing, and operational requirements before production use.

Series position: System Design and Backend - Book 04 of 14 - Study Step 19E. The local table of contents and PDF bookmarks navigate within this file; the roadmap and printed filenames navigate across the complete series.

Source provenance: curated from the independent Java SDE-2 master guide plus focused original material. Original master chapter numbers are labeled explicitly when retained in cross-references.